Terra-Scape

Integrated Sub-Surface IoT Landslide Early Warning System

Team

Geo-Scape

Target Region

Himalayan Watershed Regions

SDG Focus

SDG 11 (Sustainable Cities) & SDG 13 (Climate Action)

The Problem We're Solving

Surface-Only Sensors

Existing systems rely on surface sensors that easily trigger false alarms from rain, animals, and wind—eroding community trust through alarm fatigue.

Missing the Real Danger Zone

No monitoring occurs at the deep soil slip-plane where actual failure begins. Critical early warning signs remain invisible until it's too late.

Inaccessible Slopes

Dangerous slopes between 30°–45° remain unmonitored due to physical inaccessibility, leaving vulnerable communities unprotected.

Communication Gaps

Lack of real-time alerts in local languages prevents timely evacuation and effective emergency response in high-risk areas.

Introducing Terra-Scape

A low-cost IoT-based landslide prediction system designed to save lives through early detection at the source.

Sub-Surface Detection

Sensors buried 2 meters deep catch hidden soil changes that surface cameras completely miss—detecting instability at its origin.

The Intelligence Bridge

Combines static GIS maps with live IoT data to show exactly when danger hits, transforming raw data into actionable intelligence.

Critical Lead Time

Provides the vital early warning window needed for safer evacuation of vulnerable communities before collapse occurs.

Our Three-Layer Solution

Terra-Scape Tri-Layer Architecture integrates hardware, satellite analysis, and community intelligence.

Tier 1: Terra-Dart Node

Ground sensor installed approximately 2 meters underground to avoid surface noise, measuring deep moisture and ground movement where landslides originate.

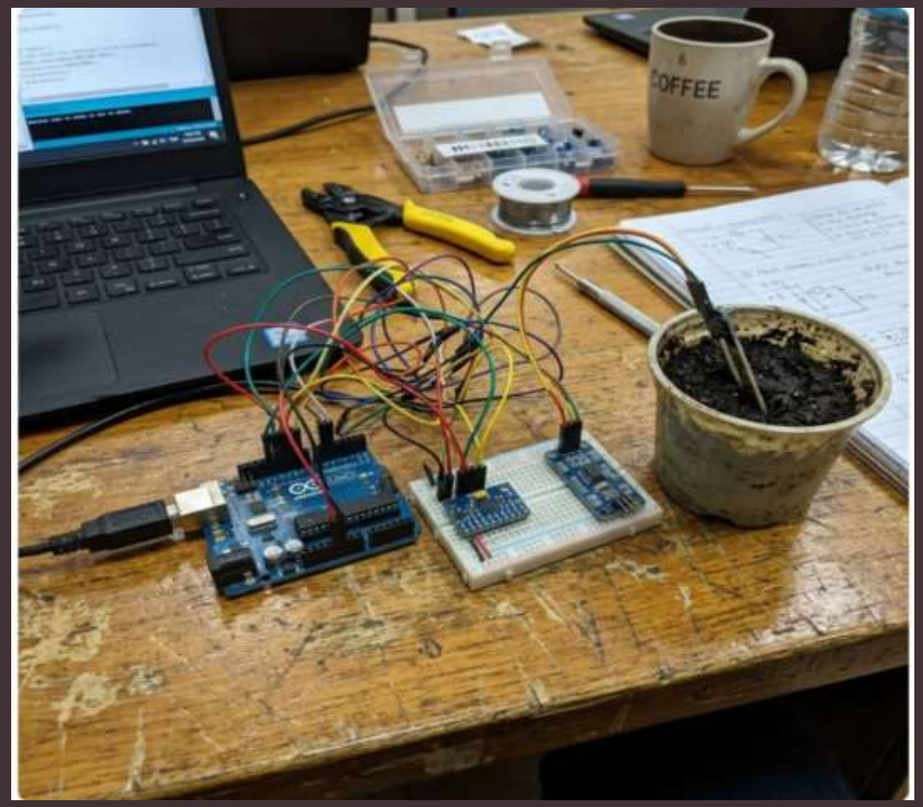
Tier 2: GIS & Satellite Layer

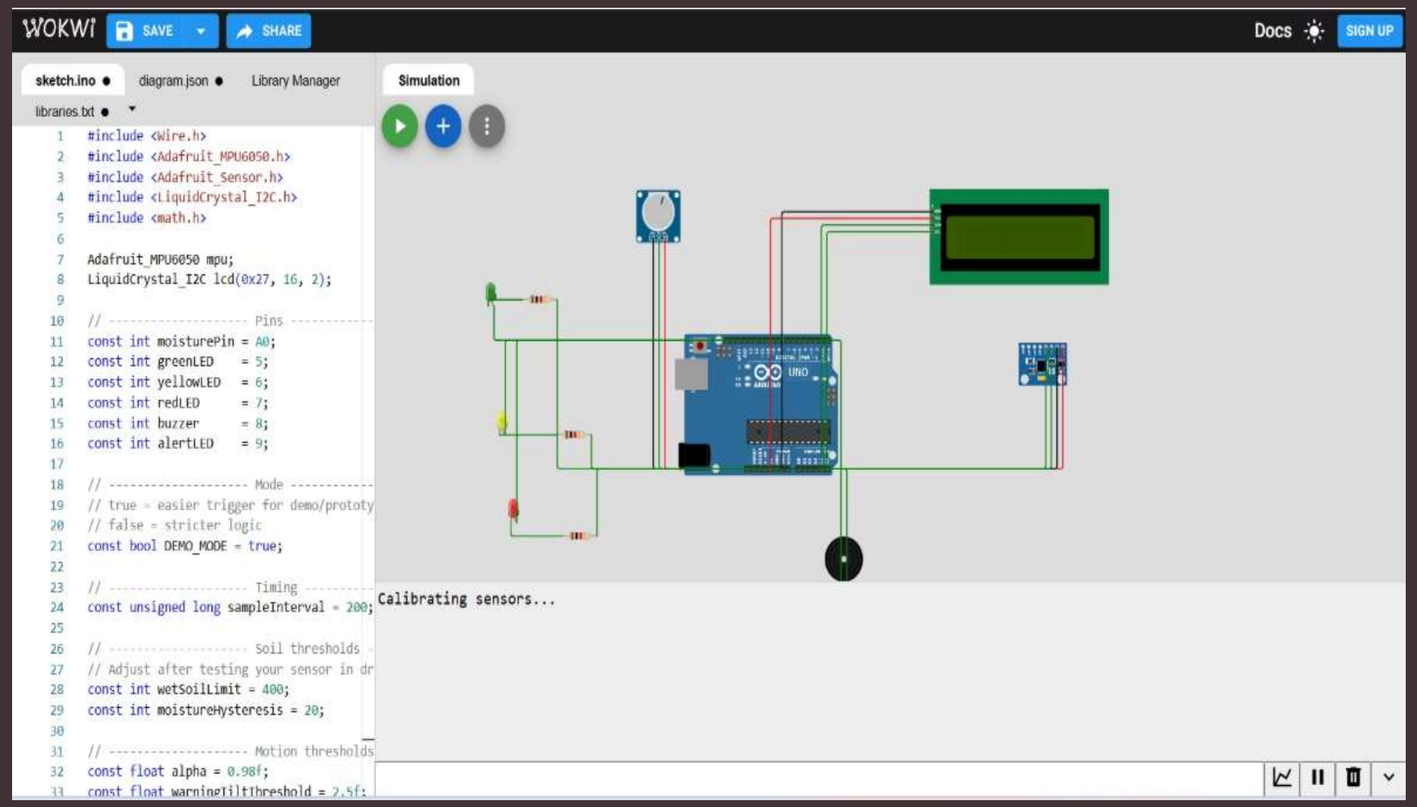
Synchronize live satellite feeds with QGIS to monitor high-risk slopes in real-time, providing constant surveillance over vulnerable communities.

Tier 3: Public Command Center

Custom AI dashboard processes live data to filter false alarms and trigger instant, community-wide alerts when instability is detected.

Prototype Development





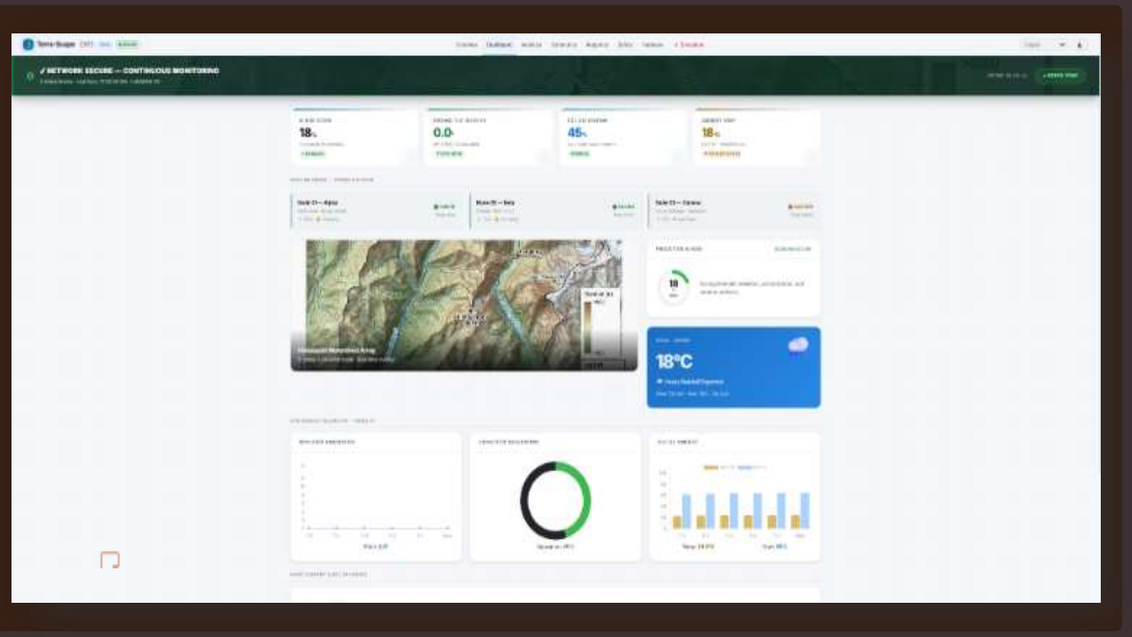
The Circuit

Custom-designed hardware integrates moisture and movement sensors with low-power wireless transmission for reliable underground operation.

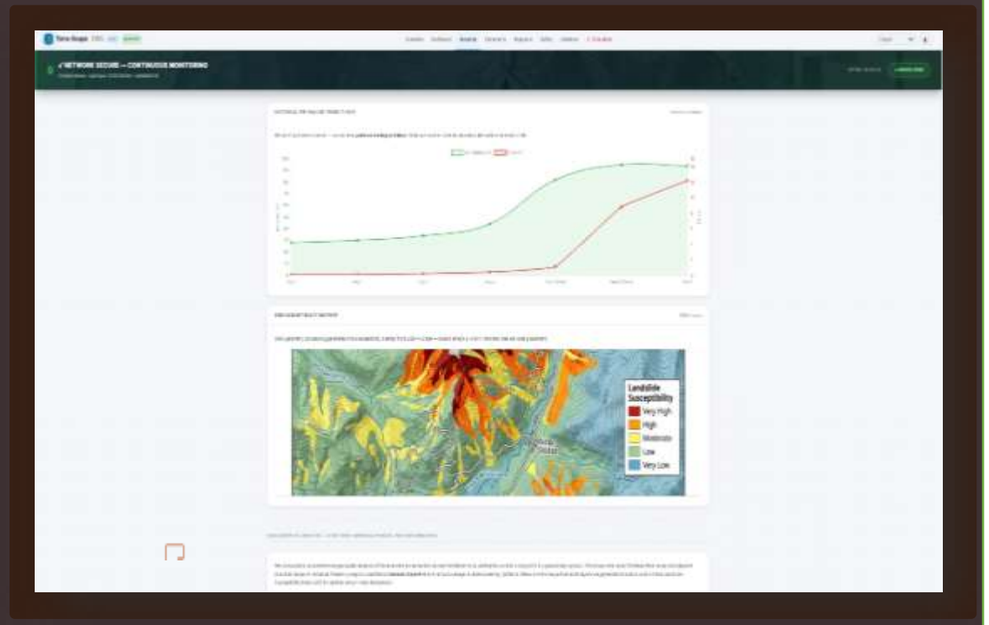
Circuit Simulation

Virtual testing ensures component reliability and power efficiency before field deployment in harsh mountain environments.

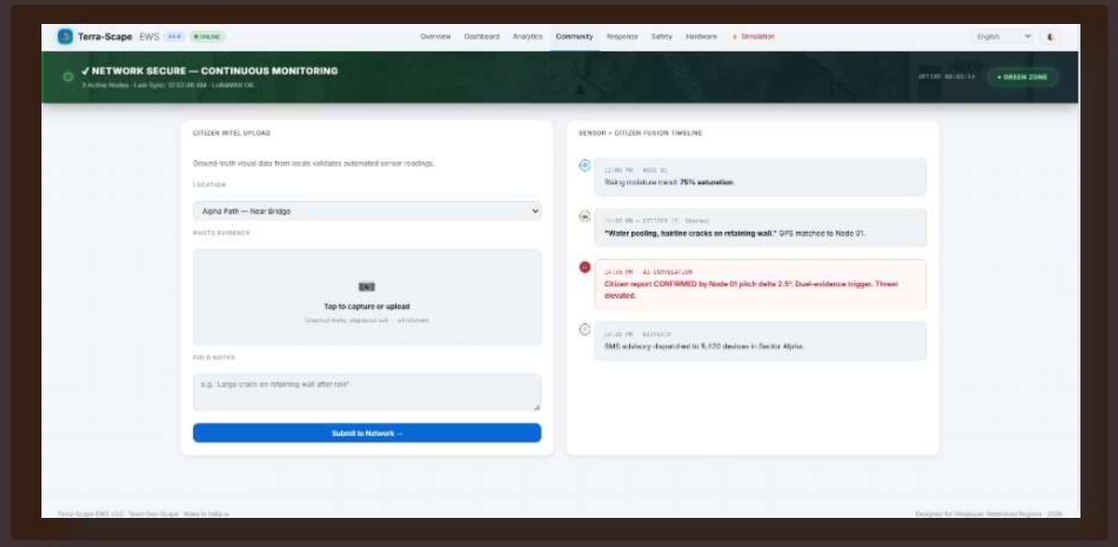
Terra-Scape Web Platform



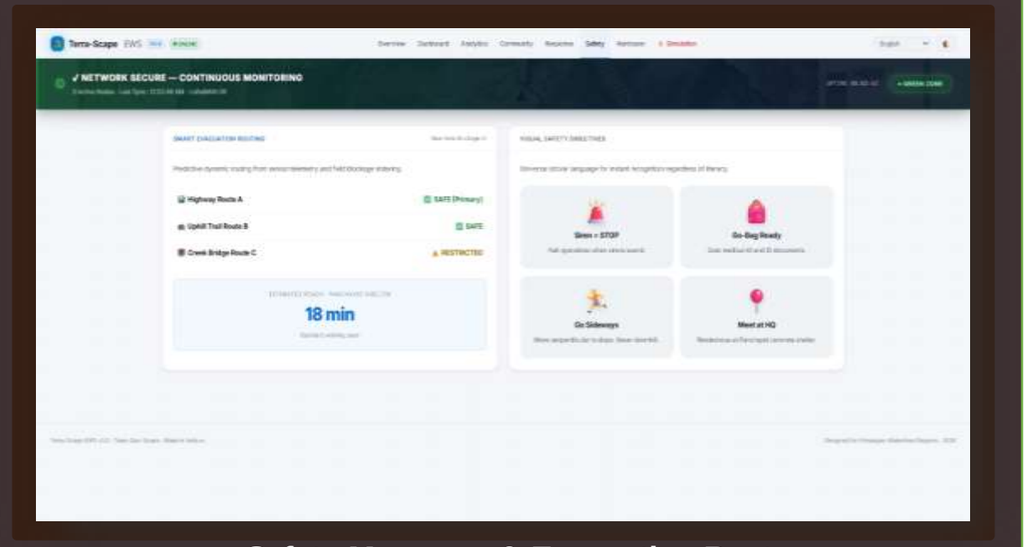
Dashboard



Real-Time Sensor Analytics



Crowdsourcing



Safety Measures & Evacuation Routes

Terra-Scape Website Features

1. Live Monitoring Dashboard

2. Real-Time Sensor Analytics

3. GIS-Based Risk Mapping

4. AI Risk Calculator

5. Community Intelligence System

6. Emergency Response System

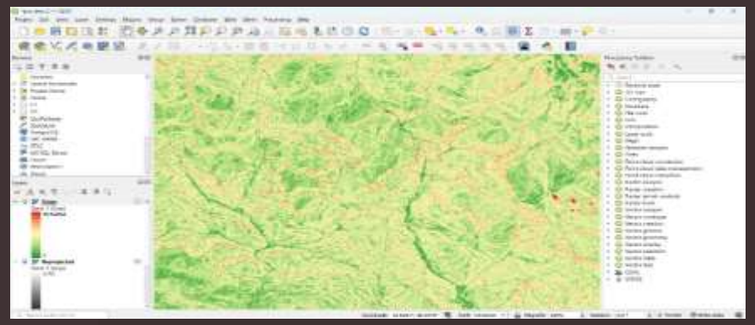
7. Alert & Broadcasting System

8. Multi-Language Support

10. Integrated Decision System

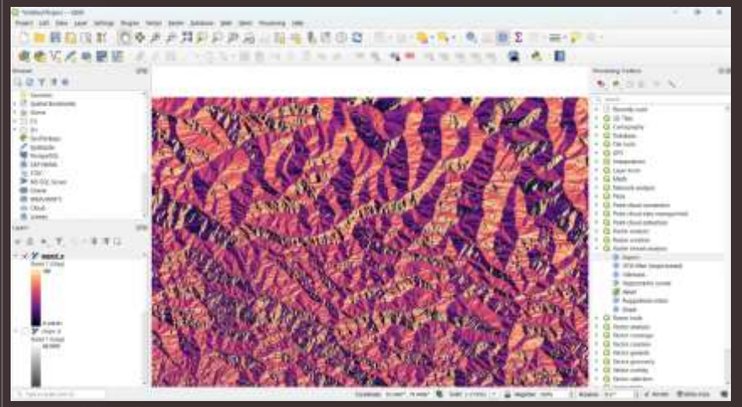
Sensors → Dashboard → AI → Alerts

QGIS Analytics Dashboard



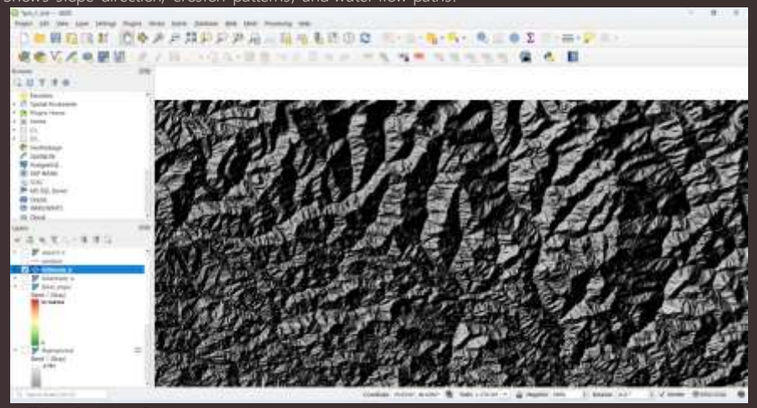
Slope Analysis

Maps 30°–45° slopes where landslides are most likely.



Aspect Mapping

Shows slope direction, erosion patterns, and water flow paths.



Hillshade Visualization

Reveals subtle terrain features hidden in flat maps.

Terra-Dart (hardware)

Terra-Dart

Weatherproof sensor probes deployed 2m deep at high-risk locations.

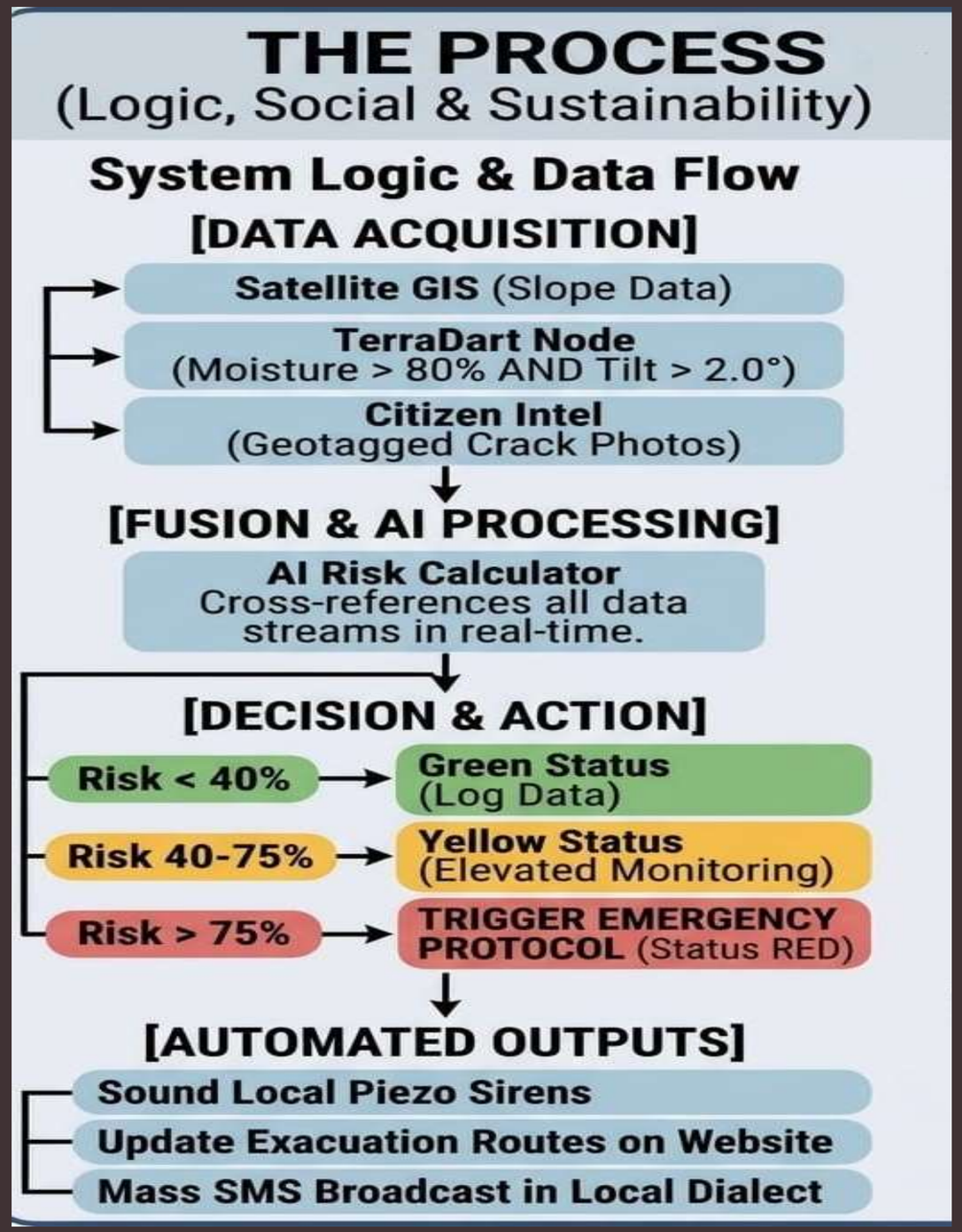
WORKING LOGIC

- **Risk Score Formula:**

- Risk = (Tilt + Moisture + GIS + Community Data)
- **Output:**

- < 40 → Safe
- 40–75 → Warning
- 75 → Danger

- Multi-source data fusion improves prediction accuracy



TERRA-SCAPE: AN INTEGRATED IoT & SUB-SURFACE LANDSLIDE EARLY WARNING SYSTEM

Bridging the gap between Himalayan terrain and human safety using Frugal IoT, GIS, and AI.

Track: Innovation Challenge | SDG 11 & 13 Focus

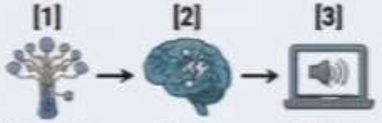
PILLAR 1:

THE CHALLENGE
(Problem Statement)

Loss of Life & Infrastructure: Himalayan communities are in persistent danger from rainfall-induced slope failures.

False Alarm Epidemic: Existing surface-level systems are triggered by wind, rain, and animals, causing communities to ignore vital warnings.

Introduction
3-Tier System



Nervous System (Sensors/ TerraDart) → **Brain** (GIS/AI processing) → **Voice** (Multi-lingual Web Command Center)

PILLAR 2:

THE TECHNOLOGY
(Hardware Layer)

Utilitarian, Reliable, and 100% Sub-Surface.

The "TerraDart" Node
(Cross-Section Diagram)

Sub-Surface Advantages

Nodes are installed 2 noders deep, innoring surface "noise" like rain and wind. Strens only trigger when the earth actually fractures at the "Slip plane.

PILLAR 3:

THE INTEL
(GIS & Community Layer)

Tier 1: Satellite GIS Mapping

Tier 2: Integrated Monitoring Website

The website serves as a public command hub, presenting aggregated risk maps and node acomality for multi-motas sider visualization. (Integration of GIS, Community data, and Node feeds).

Tier 2: Community Intelligence (Crowdsourcing)

"Report a Ground Shift" Feature: Villagers can upload geotagged photes. If a community report correlates with a sensor node anomaly, the system exponentially increases the risk score.

PILLAR 4: THE PROCESS & IMPACT

THE PROCESS
(Logic & Flow)

System Logic & Data Flow

[DATA ACQUISITION]

- **Satellite GIS** (Slope Data)
- **TerraDart Node** (Moisture > 80% AND Tilt > 2.0°)
- **Citizen Intel** (Geotagged Crack Photos)

↓

[FUSION & AI PROCESSING]

AI Risk Calculator Cross-references all data streams in real-time.

↓

[DECISION & ACTION]

- Risk < 40% → **Green Status** (Log Data)
- Risk 40-75% → **Yellow Status** (Elevated Monitoring)
- Risk < 75% → **TRIGGER EMERGENCY PROTOCOL** (Status RED)

↓

[AUTOMATED OUTPUTS]

- **Sound Local Piezo Sirens**
- **Update Exacuation Routes on Website**
- **Mass SMS Broadcast in Local Dialect**

THE IMPACT
(Social & Sustainability)

Multi-Lingual Public Command Center (Website Mockup)

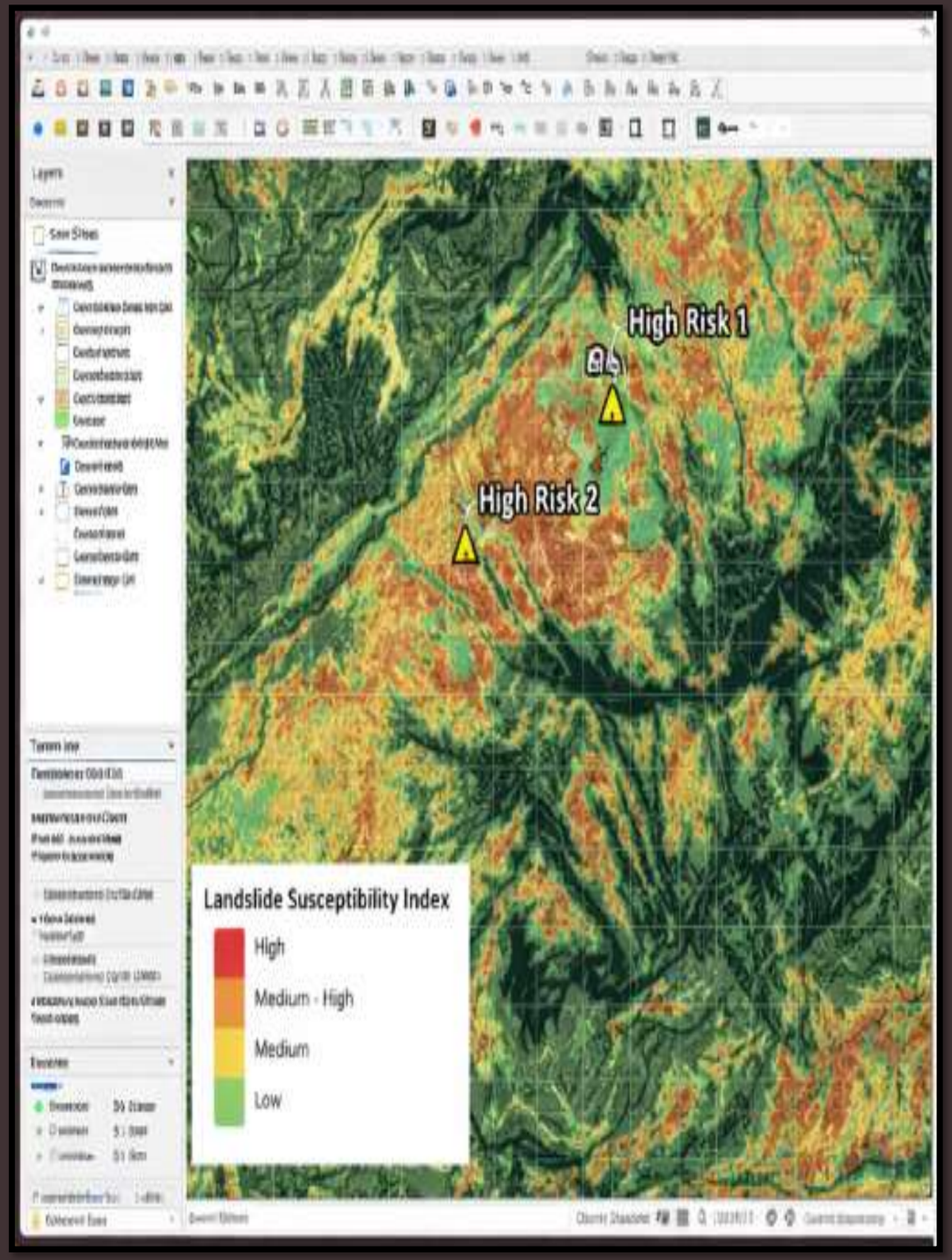
The first system designed explicity for **low-literacy, multi-dialect rural** accessibility.

Deployment Strategy & Sustainability

Villager with hand-auger (Community Jobs) | **UAV dropping nodes** (Steep Cliff Deployment)

Conclusion

95% reduction in false alarms
2.8-year battery life.
Low-cost Make-in-India solution.



Research & Precision Mapping

NASA Satellite Data

SRTM elevation models provide 30-meter resolution terrain mapping across entire watersheds.

Layer Stacking

Slope, contour, hill-shade, and vegetation data combine into comprehensive vulnerability picture.

LSI Mapping

Landslide Susceptibility Index pinpoints exact deployment locations for maximum impact.

Deep-Soil Focus

Sub-surface moisture tracking eliminates false alarms from surface rain while detecting actual failure mechanisms.

Advantages, Limitations & Deployment

Advantages

- Low-cost design enables large-scale deployment
- High accuracy through sub-surface detection
- Minimal false alarms with AI filtering
- Weatherproof construction for harsh conditions
- Community participation builds trust

Impact

- Saves lives through early warning systems
- Rebuilds trust in alert systems
- Affordable for rural and remote areas
- Minimal infrastructure requirements
- Sustainable, low-e-waste design

Limitations

- Requires initial calibration at each site
- Periodic battery replacement needed
- Internet connectivity required for full functionality

Deployment Strategy

Workflow: Map (QGIS) → Install (Terra-Dart) → Monitor (Sensors) → Predict (AI) → Alert (Users)

Methods

Manual Deployment: Local teams install sensors at identified high-risk slopes

Drone Deployment: Autonomous drones place sensors in inaccessible terrain

Scalability

- Low-cost design enables **large-scale deployment**
-
- Can be expanded to **multiple villages and regions**
-
- Integration with **cloud, mobile apps, and government systems**
- Supports **real-time monitoring at national level**

Future Plan

- Upgrade system with **wireless communication (ESP32 / LoRa)**
- Integrate **cloud database for real-time data access**
- Develop **mobile app for alerts and monitoring**
- Improve accuracy using **AI/ML prediction models**
- Deploy system in **multiple high-risk regions**
- Collaborate with **government disaster management agencies**

Conclusion

Next-Generation Early Warning

Terra-Scape combines ground sensing, satellite analysis, AI prediction, and community participation into one integrated system.

Social Impact

Saves lives, provides dynamic evacuation paths, and rebuilds trust in warning systems—directly supporting SDG 11 (Sustainable Cities).

Environmental Impact

Zero heavy concrete foundations and sustainable low-e-waste design align with SDG 13 (Climate Action) principles.

Scalable Solution

Ultimate cost-effective, scalable, and highly accurate solution for Himalayan regions and rural mountain communities worldwide.

References

•NASA SRTM 30m Digital Elevation Models (DEM) Dataset.

•QGIS Geographic Information System (Open-Source Geospatial Foundation).

•State Disaster Management Authority (SDMA) Common Alerting Protocols

THANK YOU